

Azure Machine Learning

What is Machine Learning ?

Using **known data**, develop a **model** to predict **unknown data**.

Machine Learning is an AI technique that teaches computers to learn from experience. Machine learning algorithms use computational methods to “learn” information directly from data without relying on a predetermined equation as a model.

Text Analytics: User reviews

Positive

Text Analytics - Preview by [Azure Machine Learning](#) Demo Sample Code Doc

Try out the Azure ML [Text Analytics service](#) - for free.

To use the service in production, you can get access to its API by [signing up](#) for it (click on "Sign Up" on the right pane on that page). For questions/comments, please use the "Feedback" button in the top right.

I love this presentation. I found it very useful.

Analyze

Sentiment:

98 %

Key phrases highlighted below:

I love this **presentation**. I found it very useful.

Negative

Text Analytics - Preview by [Azure Machine Learning](#) Demo S

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I hate this presentation. I found it very useless.

Analyze

Sentiment:

7 %

Key phrases highlighted below:

I hate this **presentation**. I found it very useless.

What is Machine Learning ?

Using **known data**, develop a **model** to predict **unknown data**.

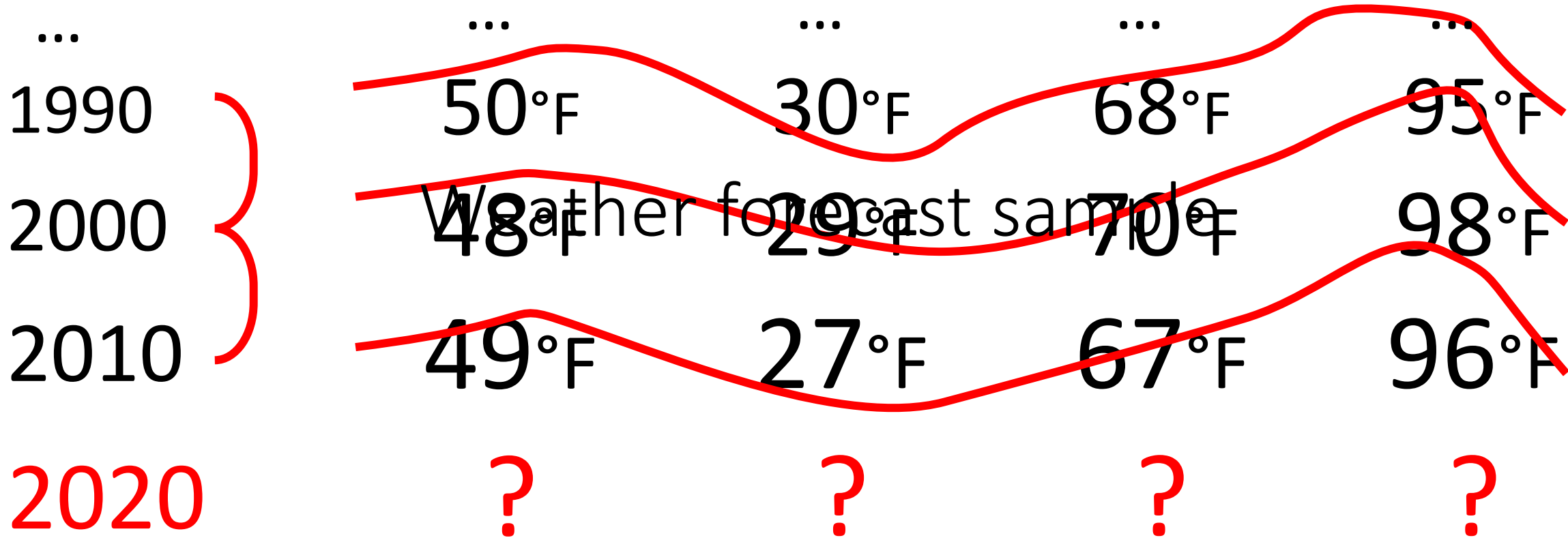
Known Data: Big enough archive, previous observations, past data

Unknown Data: Missing, Unseen, not existing, future data

Model: Known data + Algorithms (ML algorithms)

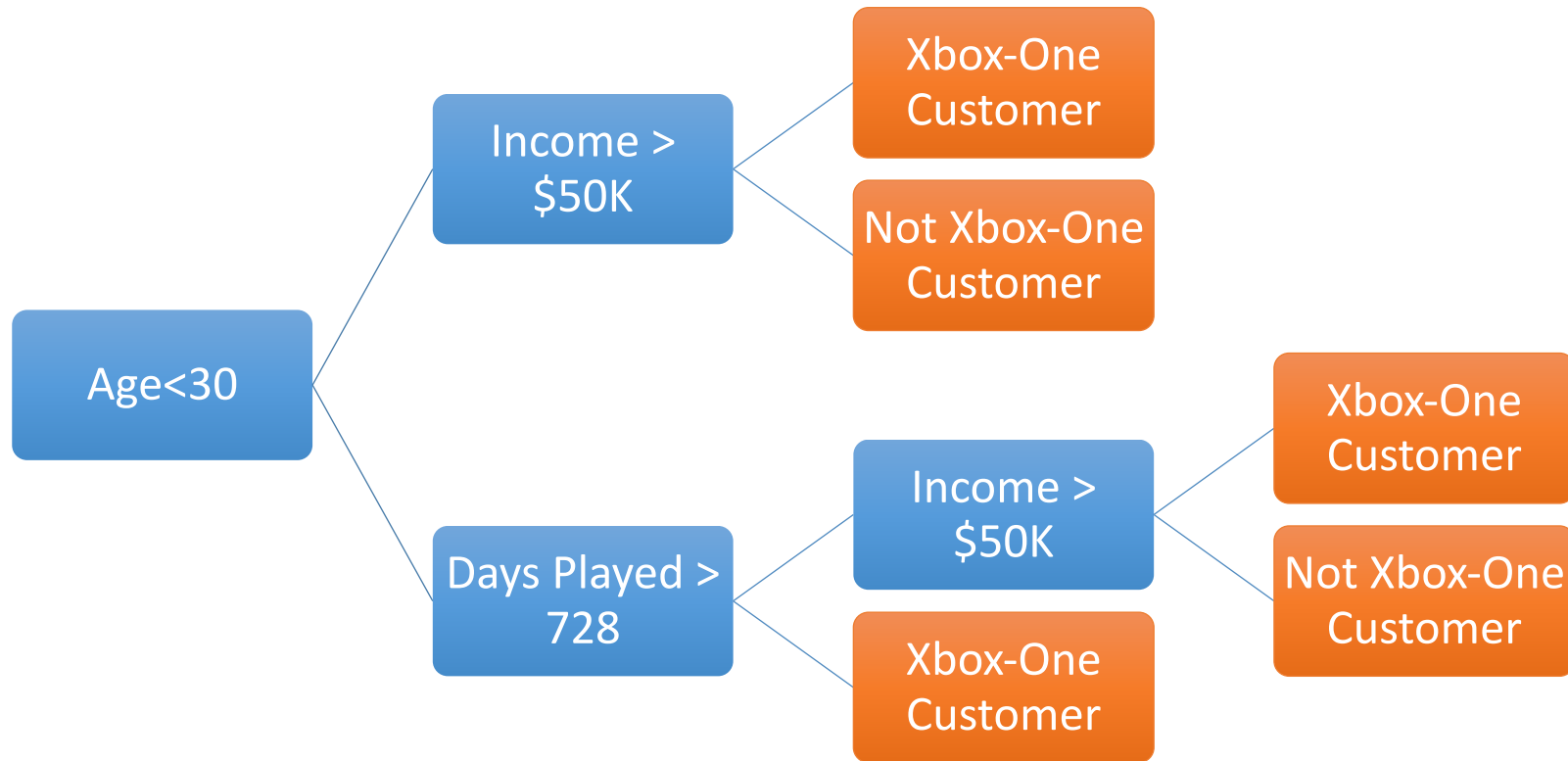
EXAMPLE

Known data
Model
Unknown data



EXAMPLE

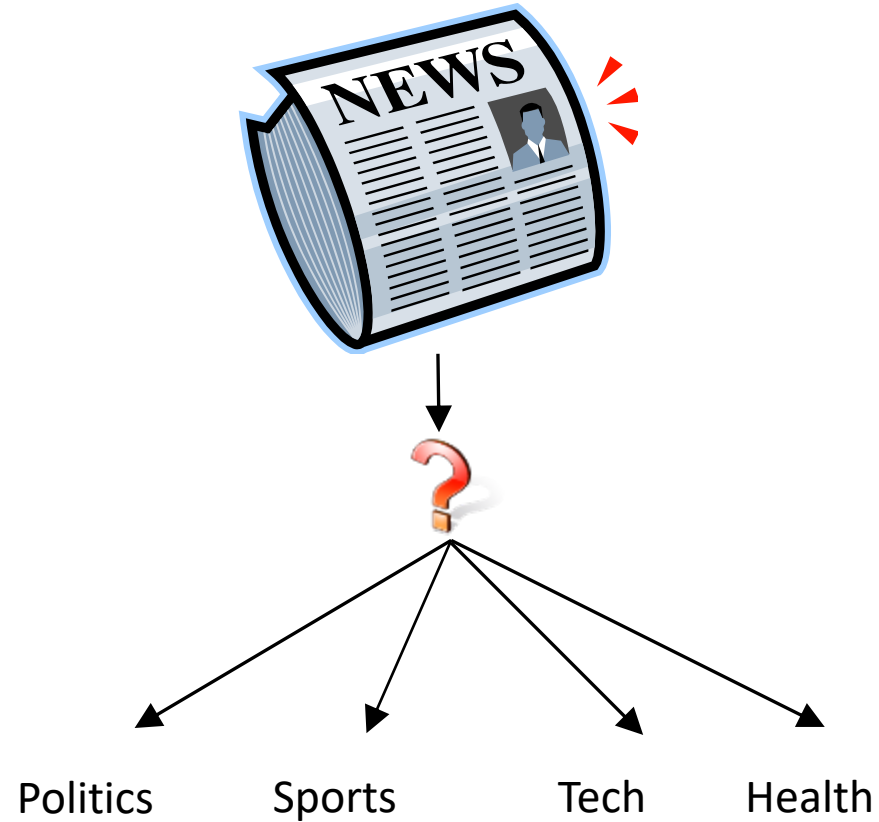
Model (Decision Tree)



EXAMPLE

Model (Classification)

Classify a news article as (politics, sports, technology, health, ...)



Known data (Training data)

Documents Labels



Tech



Health



Politics



Politics



Sports

Documents consist of unstructured text. Machine learning typically assumes a more structured format of examples

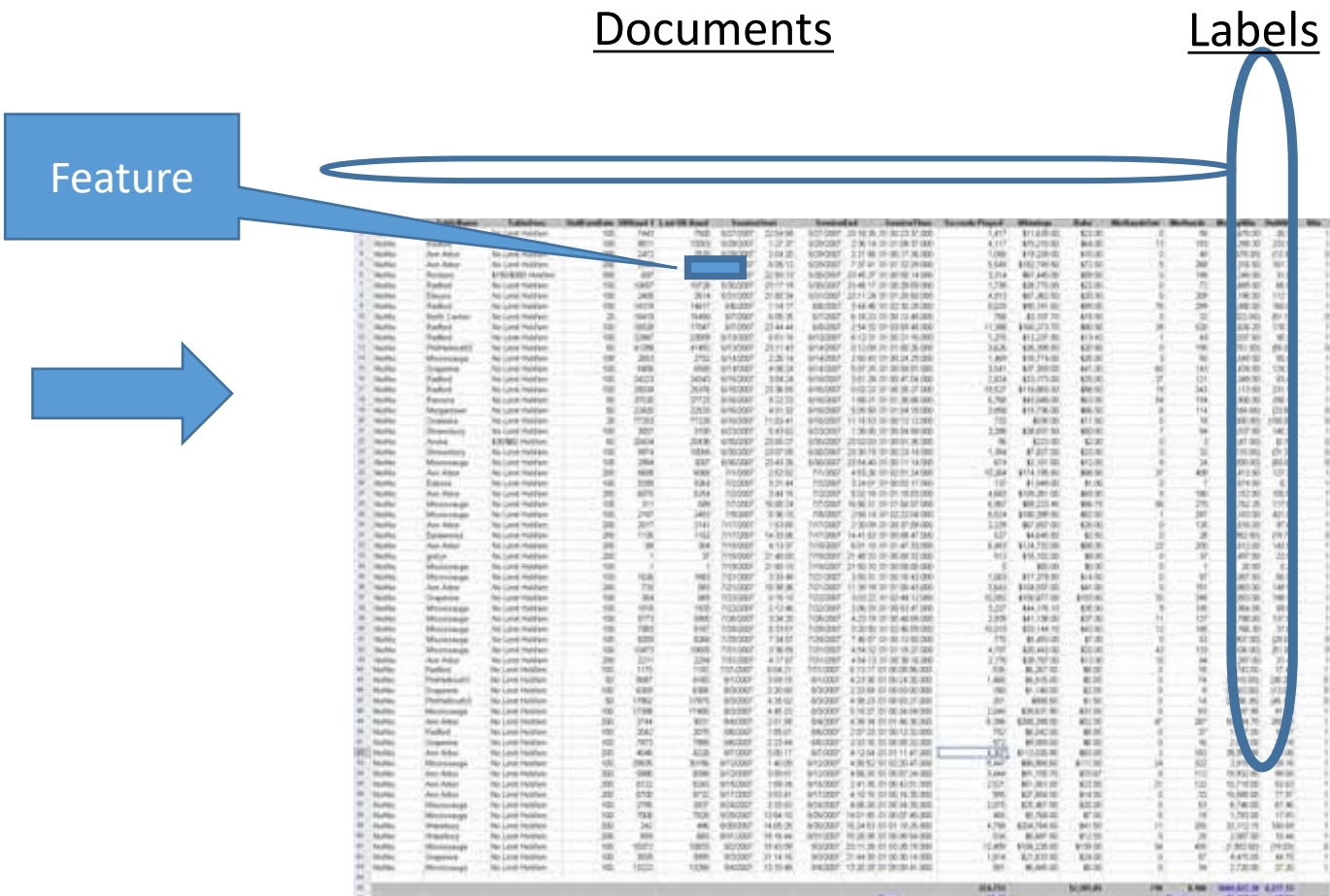
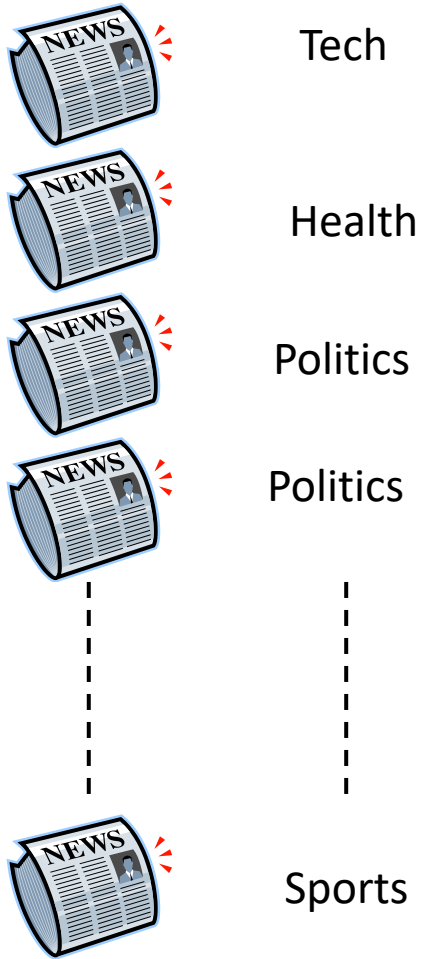
Process the raw data

Using **known data**, develop a **model** to predict **unknown data**.

Known data (Training data)

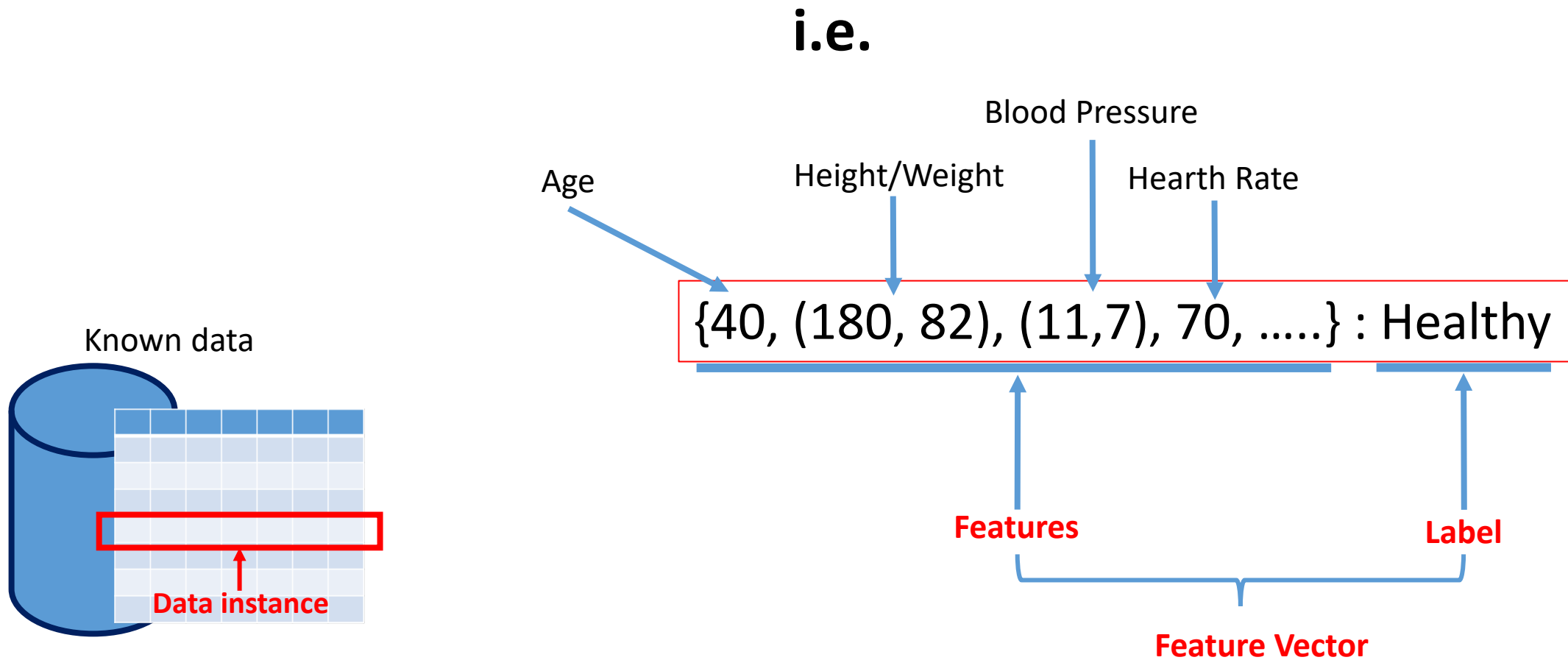
Documents Labels

Process each data instance to represent it as a feature vector



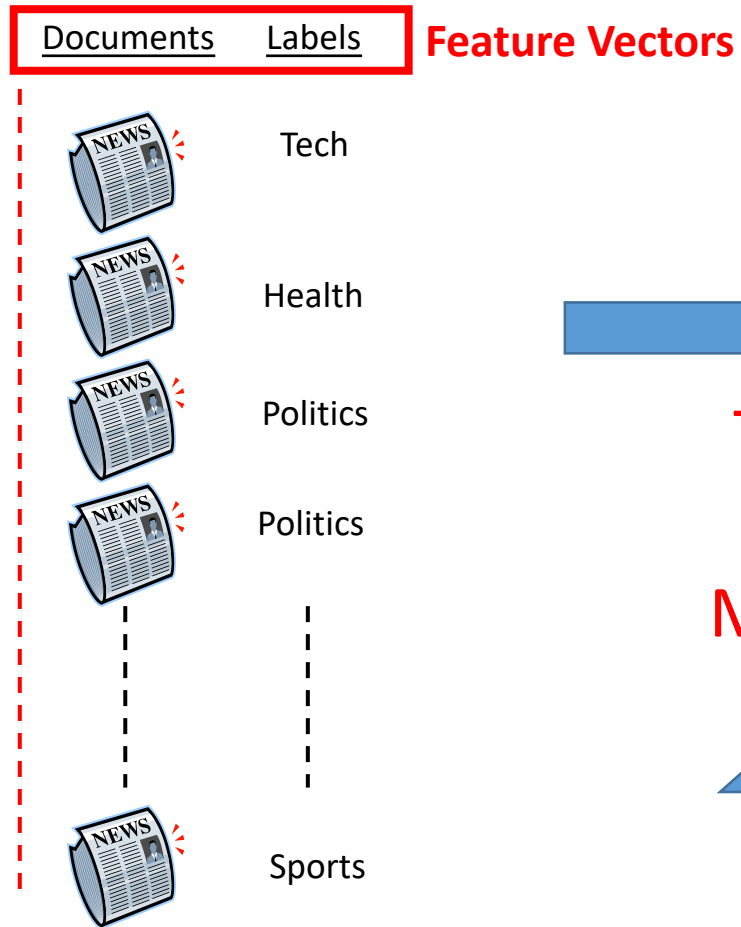
Using known data, develop a model to predict unknown data.

Feature vector



Developing a Model

Training data



Train
the
Model



Base
Model

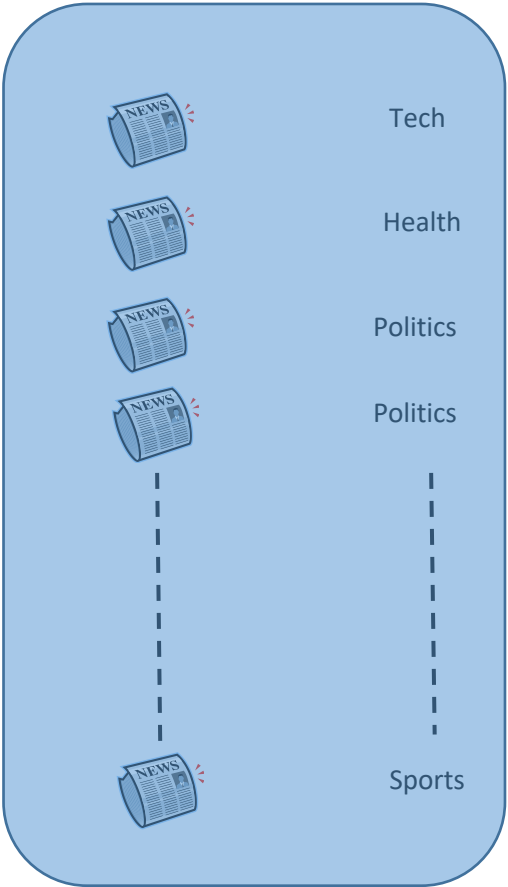


Adjust
Parameters

Using **known data**, develop a **model** to predict **unknown data**.

Model's Performance

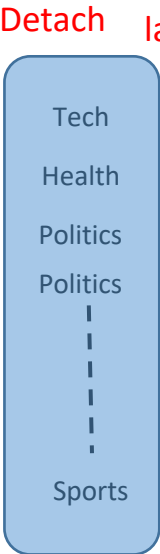
Known data with **true labels**



Split

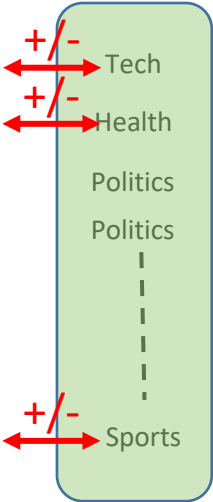
Test data 20%

Training data 80%



True labels

Predicted labels



Test trained model with features

Compare prediction with true labels

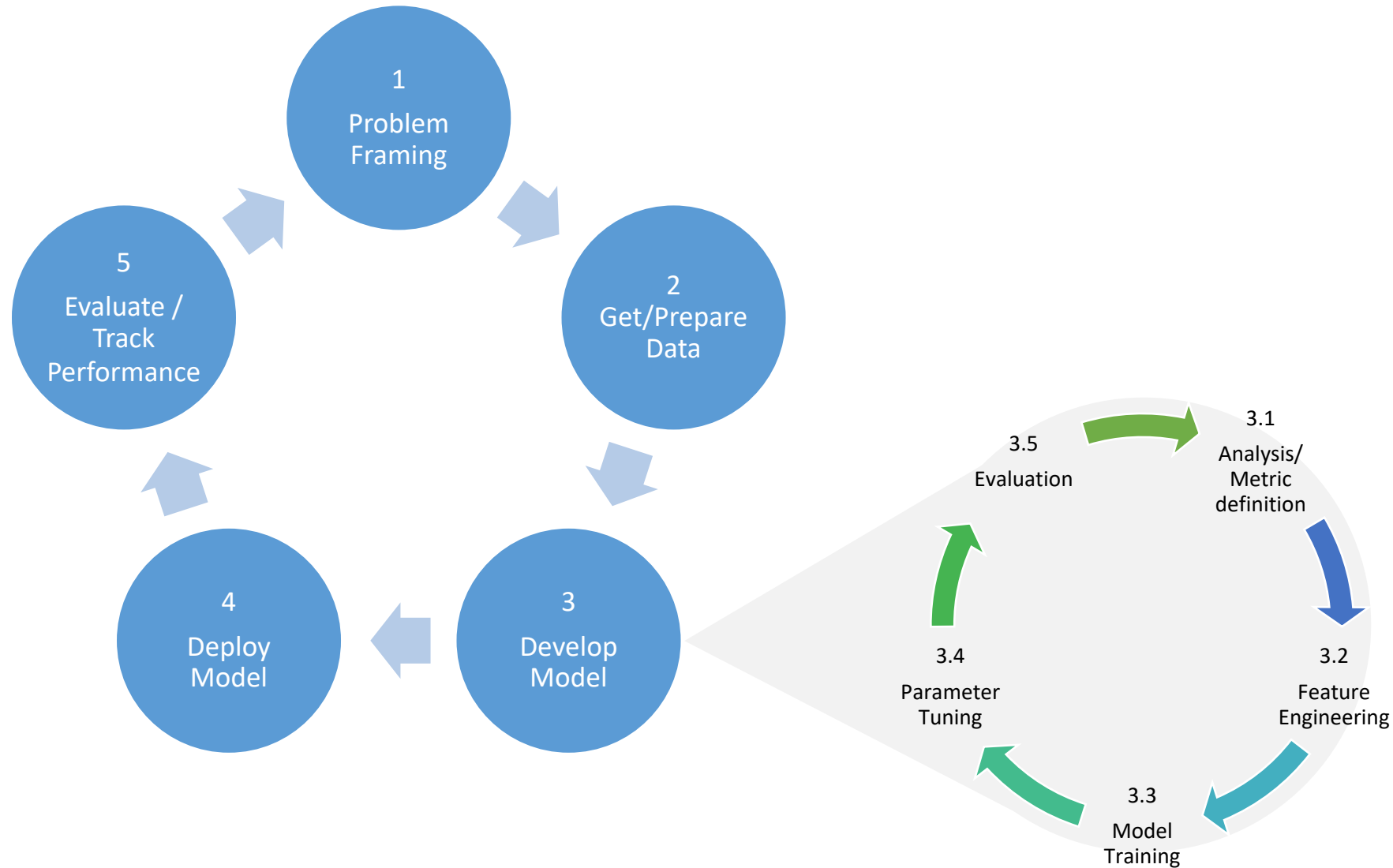
Train the Model



Model's Performance

Difference between "True Labels" and "Predicted Labels"

Steps to Build a Machine Learning Solution



Example use cases

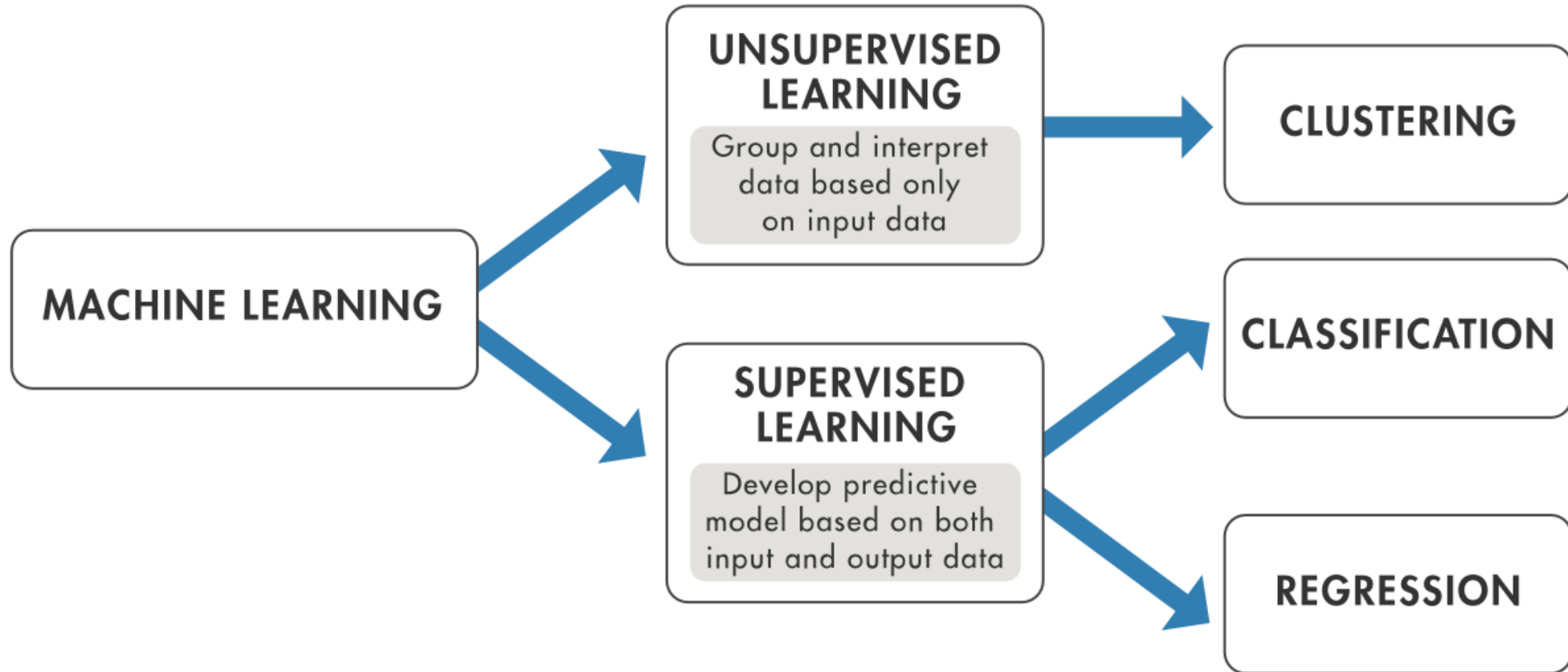
 Finance and risk	 Sales and marketing	 Customer and channel	 Operations and workforce
 Revenue Forecasting	 Sales forecasting	 User segmentation	 Agent allocation
 Portfolio optimization	 Demand forecasting	 Personalized offers	 Warehouse efficiency
 Investment modelling	 Sales lead scoring	 Product recommendation	 Smart buildings
 Fraud detection	 Marketing mix optimization		 Predictive maintenance
 Risk management			 Supply chain optimization

Machine Learning Algorithms

- ML Algorithm defines how your **model** will react
- Which Algorithm to use? Depends on:
 - Data Quality
 - Data Size
 - What you want to predict
 - Time constraint
 - Computation power
 - Memory limits



Machine Learning Algorithms



Machine Learning Algorithms

Two major category of algorithms

- **Supervised** - takes a known set of input data and known responses to the data (output) and trains a model to generate reasonable predictions for the response to new data
- **Unsupervised**- Unsupervised learning finds hidden patterns or intrinsic structures in data. It is used to draw inferences from datasets consisting of input data without labeled responses.

Most commonly used machine learning algorithms are **supervised** (requires labels)

• **Supervised** learning examples

- Email is genuine or spam
- Weather forecasting
- Tumor is cancerous or benign

• **Unsupervised** learning examples

- These customers are similar – marketing applications
- Identifying frauds – insurance applications

Why you need to know these algorithms?

- If you want to answer a YES|NO question, it is **classification**
- If you want to predict a numerical value, it is **regression**
- If you want to group data into similar observations, it is **clustering**
- If you want to recommend an item, it is **recommender system**
- If you want to find anomalies in a group, it is **anomaly detection**

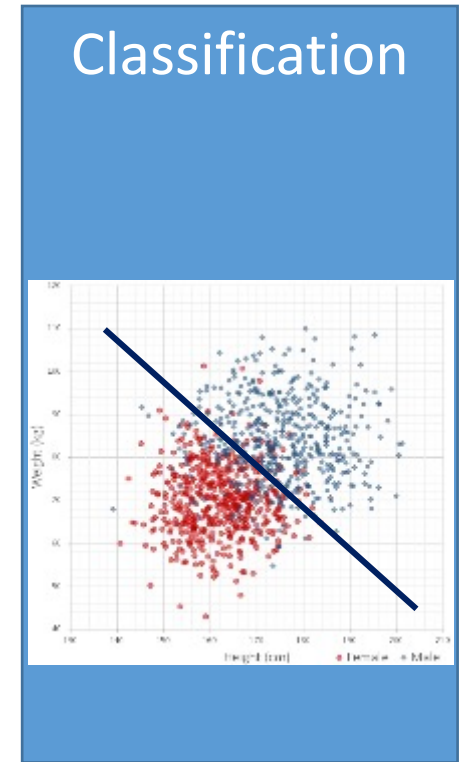
and many other ML algorithms for specific problem

Classification

Scenarios:

- Which customer are more likely to buy, stay, leave (churn analysis)
- Which transactions|actions are fraudulent
- Which quotes are more likely to become orders
- Recognition of patterns: speech, speaker, image, movement, etc.

Algorithms: Boosted Decision Tree, Decision Forest, Decision Jungle, Logistic Regression, SVM, ANN, etc.

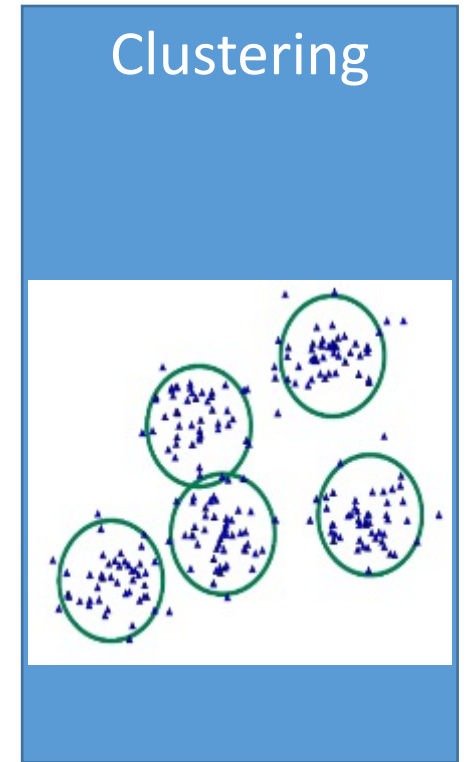


Clustering

Scenarios:

- Customer segmentation: divide a customer base into groups of individuals that are similar in specific ways relevant to marketing, such as age, gender, interests, spending habits, etc.
- Market segmentation
- Quantization of all sorts, such as, data compression, color reduction, etc.
- Pattern recognition

Algorithms: K-means

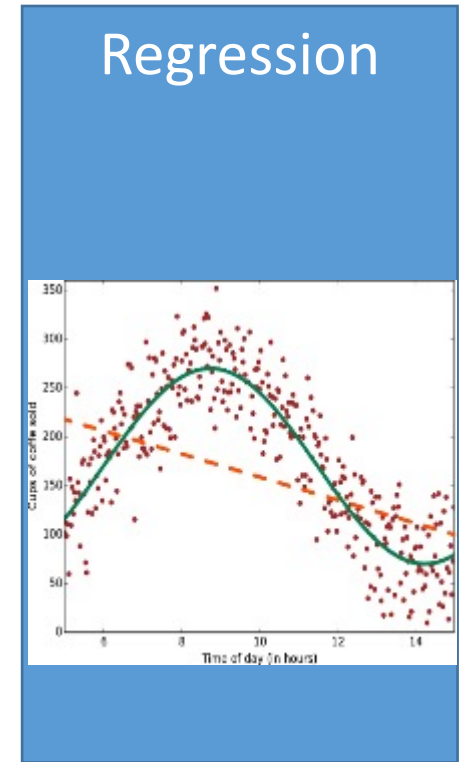


Regression

Scenarios:

- Stock prices prediction
- Sales forecasts
- Premiums on insurance based on different factors
- Quality control: number of complaints over time based on product specs, utilization, etc.
- Workforce prediction
- Workload prediction

Algorithms: Bayesian Linear, Linear Regression, Ordinal Regression, ANN, Boosted Decision Tree, Decision Forest



Regression versus Classification

Does your customer want to predict | estimate a number (regression) or apply a label | categorize (classification)?

- Regression problems

- Estimate household power consumption
- Estimate customer's income



- Classification problems

- Power station will | will not meet demand
- Customer will respond to advertising

